BTS-V(S)-04.18-0762 Reg. No. [ ] **A**

B.Tech. Degree V Semester Supplementary Examination April 2018

CS 15-1504 OPERATING SYSTEM

(*2015 Scheme*)

Time : 3 Hours Maximum Marks : 60

PART A

(Answer *ALL* questions)

(10 × 2 = 20)

I. (a) What is a system call? Give an example.

(b) What is a process? With a neat diagram write about various states of a process.

(c) What is meant by guaranteed scheduling? Explain.

(d) What is meant by inverted page table? Compare it with normal page table.

(e) Differentiate between paging and segmentation.

(f) What is meant by deadlock? What are the necessary conditions for a deadlock to occur?

(g) What is meant by non resource deadlock? Explain with an example.

(h) Explain briefly interrupt handlers.

(i) Write briefly about file system implementation in Linux operating system.

(j) Discuss about the applications of real time systems.

PART B

(4 × 10 = 40)

II. (a) Explain inter process communication. (5)

(b) What is critical section problem? Explain the use of semaphores in solving the producer consumer problem. (5)

OR

III. (a) Explain with an example any two CPU scheduling algorithms. (5)

(b) Write notes on (i) Context switch and (ii) two level scheduling. (5)

IV. (a) Explain the following terms related to memory management. (5)

(i) Swapping.

(ii) Compaction.

(b) Explain about any two memory management mechanisms. (5)

OR

V. (a) Explain with an example how address mapping is done in paging. (5)

(b) What is the need for page replacement? Explain any two page replacement algorithms with an example. (5)

VI. (a) Explain various deadlock prevention mechanisms. (5)

(b) Explain deadlock detection in a system with resources having single instance. (5)

OR

VII. Explain with an example the Banker's algorithm for deadlock avoidance in a system with resources having multiple instances. (10)

VIII. (a) Explain with an example any two disk scheduling algorithms. (5)

(b) Explain briefly I/O buffering. (5)

OR

IX. (a) Explain scheduling in RTOS. (5)

(b) Explain the characteristics of a real time operating system. (5)

***